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MAJOR CONTRIBUTORS OF THEIR VALUABLE INPUTS



Anool Mahidharia
director,
Lumetronics



Boris Landoni
technical director
and co-founder,
www.open-electronics.org



Dhaval Vasa
solution architect,
einfochips



Kavita Arora
founder, Bangalore
MakeSpace and
Open Source
Creativity (BMOSC)
Community



Priya Kuber
director, Arduino
India



Varun Sivaram
CEO and founder,
Technigrit
Systems Pvt Ltd

so, informs Boris Landoni, technical director and co-founder, www.open-electronics.org. He says, "One had to hack a product to figure it out, but now they help us, give us ready pieces of code, enabling us to build better." For instance, he adds, "After Microsoft's Kinect was hacked, Microsoft had seen a surge in sales and has come up with a software development kit to use it. People have started working on creating projects of which Microsoft certainly had not thought of."

New innovations

With the advent of 3D printing, laser cutting and computer numerical control (CNC) machining (and open source machines for these techniques), people with bright ideas are finding it easy to build things. "I foresee a lot of other manufacturing related innovations which will make it even more easier for anyone to create open hardware. Small, open source pick-and-place machines, open source laptops and open source semiconductors (Parallax, Momentum) are a few examples," shares Anool Mahidharia, director, Lumetronics.

The world of open source hardware runs fast, but this year there has been an explosion of 3D printers. All are strictly open source but with some additional features than those presented before. "This is also the year of Linux," believes Landoni. He adds, "After Raspberry Pi has been on the market, new embedded boards integrate increasingly powerful processors and the features are more interesting."

Interesting initiatives

Almost every large-scale industrial production process is now available in small format. This allows makers to develop smart, innovative, open hardware for prototyping. With a prototype in hand, makers can then pitch for funding via crowd sourcing websites. This completely bypasses the traditional Manufacturer - Distributor - Reseller - Retail model.

The distribution chain is also shortening. Who sells also produces, and often with very low margins. Like Raspberry Pi and Arduino, there is not much space for the retailer, to the benefit of the consumer.

Many applications will run on the Web; the open source codes can be edited and compiled directly on line for better development time and reality. "This is not a futuristic vision, already there are these realities, but will grow like wildfire and will be more user friendly," notes Landoni.

The road ahead

With all these initiatives and developments, the aim to create an ecosystem of open source hardware in India looks brighter. With smaller sensors, smaller actuators, high energy-density batteries, low-power microcontrollers and connected devices emerging, the 'Internet of Things' revolution is also around the corner. Cost of boards and tools will become more affordable, and there will be a lot more participation from people. This will enable everyone to translate their thoughts and ideas into prototypes efficiently and effectively. ●

The author is a senior technical correspondent at EBY